

The first business

You just learned from the university homepage that one of the snack bars has its rental period finishing at the end of 2021 so there will be a tender process to choose the new operator. Just as formerly, a rental period of five years will be offered and the highest bidder will be chosen. (Everyone has to adapt to some limitations on product offering.) As you do not yet know how to go on after the university, you decide to team up with your friends and try run the snack bar. Anyway, you have the serious advantage of knowing of your own experience what should be changed and what students wish for.

Based on your estimations, HUF 6 Million is needed to start the business. The parents of one of you have considerable experience in running pubs and canteens, so they can help you with reasonable estimations on operation and risks. After a step-by-step improvement of the initial offering, you expect to leave the business after five years and look for a professional career. When you quit, the invested and current assets can be sold to the new owner at 80 percent of the book value at the beginning of 2027.

The corporate tax is predicted to remain stable at 10% during all the planning horizon. At the start-up, you can not count on a bank loan, but one of your friends is able to get some family loan at a rate of 15 percent – just in line with the risk. This loan will be gradually replaced by less expensive sources, while debt will always amount to 30% of the invested capital. (Debt will be reset at the end of each year.) You expect a sales of HUF 20 million for the first year.

	2022	2023	2024	2025	2026
Growth of Sales		12.00%	10.00%	8.00%	6.00%
EBIT/Sales	-2.00%	10.00%	16.00%	20.00%	20.00%
Additional IC needed (K HUF)	500.00	400.00	300.00	200.00	100.00
r_D	15.00%	12.00%	10.00%	10.00%	10.00%
r_A	20.00%	17.00%	15.00%	15.00%	15.00%

1. Predict the financial statements of the firm.
2. Prepare DCF-based valuation using any methods you know of.
3. What would be the maximum rental fee worth to pay?
4. How would the maximum fee change if you pay for the rental right not as an individual, rather as firm allowing you to capitalise the right in your books? (The useful life is 5 years long.) (Note, that rental rights can not be used as a collateral for debt, thus it should not be considered when calculating the available amount of debt.)
5. What would be the maximum annual fee if you had to pay as a firm a flat fee and expense that each year?

The first business II.

You still consider to run one of the university snack bars for 5 years. You just bought the 5-year right for operation for 4 M HUF. (The purchase price will be amortised across five years in equal instalments. For the plan, you need an additional 6 M HUF start-up capital. At the end of the rental period, in 2027, the new tenant will buy all your assets and working capital at 80 percent of the book value. You expect the corporate tax rate to remain 10 percent across the whole planning horizon, the predicted sales of the first year is 20 M HUF. Contrary to the earlier plans, now you plan to finance the firm so that at the end of each year debt will be rebalance to equal to 30 percent of the business value (V).

	2022	2023	2024	2025	2026
Growth of Sales		12,00%	10,00%	8,00%	6,00%
EBIT/Sales	-2,00%	10,00%	16,00%	20,00%	20,00%
Additional IC needed	500,00	400,00	300,00	200,00	100,00
Corporate tax rate	10,00%	10,00%	10,00%	10,00%	10,00%
r_D	15,00%	12,00%	10,00%	10,00%	10,00%
r_A	20,00%	17,00%	15,00%	15,00%	15,00%

1. Predict the financial statements of the firm.
2. Prepare DCF-based valuations using as many models as possible.
3. One of your friends offers to join to the founders, but he has no money to invest. Instead, he offers his experience in supervising the employees. His work would push EBIT/Sales ratios by 2 percentage points upwards (after deducting his wage) thanks to stronger motivation and less room for “private businesses”. What is the maximum stake of equity you would offer him in exchange of this unique contribution in kind?

Forever and after

You have to plan the outlook of a future telling firm to be funded with 10 M HUF as capital. Taking profit of the abilities of the company, you joined for séance with the Black Virgin and asked her how the future numbers may look like. Using a crystal ball, she foretold the following numbers without any further explanation.

	2022	2023	2024
ROIC	20.00%	18.00%	16.00%
K	120.00%	80.00%	40.00%
r_A	12.00%	12.00%	12.00%
r_D	8.00%	8.00%	8.00%

At that moment, unfortunately, your phone started to ring. The trance broke and the Black Virgin got very angry. Luckily, you could jump away from her curse thrown on you but the door got badly burned. Thus, it is you who has to collect the missing data. Based on the assumptions of the top management after 2024 in the long run you may expect a growth of 5 percent. The operational cost of capital and cost of debt will remain unchanged at the 2024 level. The financing policy dictates to cover 40 percent of the business value from debt. The tax rate remains stable at 20 percent.

1. Is it worth to launch the firm if we expect the new investments not generate any value during the terminal value period? What will be the dividend payment ratio like during the coming years?
2. How would the plan change if we assume that there will be no value generated at all during the terminal value period?
3. How would your answer for the first two questions change once the long-term growth would amount to 6 percent?

The Financing Dilemma

You just got a position at the Ferrum Ltd., a medium sized Hungarian machinery par manufacturer. You have to fulfil both financial and accounting tasks and you have to provide support to the CEO. He asks you to review the two debt offers received for a new project asking for 1000 million HUF of capital.

Banks got a detailed financial plan the key data of which appear in the table below. The attached description highlights that after six years of operation the assets incorporating the invested capital that time can be sold mainly due to the inflation at 110 percent of their book value. Corporate tax rate will remain stable at 10 percent.

	2022	2023	2024	2025	2026	2027
ΔIC	30.00	40.00	50.00	70.00	40.00	30.00
ROIC	17.00%	16.00%	15.00%	14.00%	13.00%	12.00%

The offer of Alfa Bank includes a debt of 400 million HUF at an interest rate of 5.2 percent. The loan should be repaid in four equal instalments during the last four yours of the project before selling the assets. As a collateral, the firm would have to maintain a Debt Service Reserve Account (DSRA) with balance of 25 percent of the yearly debt service at the beginning of each year.

The Bingo Development Bank, where a good friend of the CEO works, offered a credit of 500 million HUF at an interest of 6.0 percent that is to be repaid at the end of the project. Here the DSRA requirement is 50 percent of the yearly debt service.

First, you have to estimate the operative cost of capital for the firm, for what you collected the data shown in the next table. Market premiums are always relevant for the given markets and small firm premium was estimated based on data from companies of similar size of Ferrum Lfd.

	Hungary	Germany	USA
Risk free rate	3,50%	2,50%	2,00%
Inflation	3,00%	2,00%	1,80%
β_A	1,25	1,10	0,95
Market risk premium	6,00%	5,00%	4,00%
Small firm premium	3,00%	4,00%	5,00%

1. Estimate the operative (unleveraged or industry) cost of capital (r_A) of Ferrum Ltd. in various ways.
2. What estimation challenges would you face in the real life?
3. Prepare a valuation using both APV and FCFE methods for both debt versions.
4. Which of the debts is more advantageous for the firm?

Sweet Home Furniture

Sweet Home Furniture Ltd. was founded 15 years ago. The company founded by four individuals originally focused only on furniture retail in a rented storage building. Later, they built an own department store with a significant storage facility attached.

Three years ago, the firm also launched a small manufacturing division specialised particularly on kitchen furniture and wardrobes in a nearby rented building. In the long run, they also plan to introduce upholstered furniture into their product portfolio.

It is exactly because of serious debates among owners about the future strategy to follow that two of them wish to leave the firm. Thus, a valuation is need. The other two owners (father and son, both active in the current management) hold together a stake of 60 percent wish to buy out the others. The sellers wish to prepare a multiple-based valuation as it would be the buyers who would provide the inputs for a cash flow-based valuation.

When analysing the markets, you figured out that there are several furniture manufacturers listed on nearby stock exchanges but no stores. Luckily, you could collect some information on recent transactions with corporate value, sales, and growth data. You also managed to separate financial statement according business line with the help of the chief accountant. You also discovered that from 2021 on both business units will face an effective tax burden of 10 percent as tax reductions linked to new employments in the manufacturing unit will expire.

2020 (000 HUF)	Holding	Manufacturing	Retail
Invested assets	137 000	12 000	125 000
Inventory	48 200	5 200	43 000
Receivables	15 200	1 200	14 000
Cash	19 000	3 000	16 000
Short term liabilities	19 500	4 500	15 000
Debt	85 000	5 000	80 000
Equity	114 900	11 900	103 000
Sales	228 000	38 000	190 000
EBIT	44 000	8 000	36 000
Profit from financial activities	- 8 000	- 500	- 7 500
Profit before tax	36 000	7 500	28 500
Profit after tax	32 850	7 200	25 650
Expected yearly growth		20%	5%

Company name	Nationality	Main activity	Forward P/E	g
Bútor Co.	Hungarian	Furniture manufacturing	6.0	7%
Meble SA	Polish	Furniture manufacturing	9.0	5%
Mobile SA	Romanian	Furniture manufacturing	14.0	21%
Möbel AG	German	Furniture parts manufacturing	11.0	15%
			EV/S	g
Couch Partnership	Hungarian	Furniture retail	3.1	6%
Otthon Ltd.	Hungarian	Furniture retail	2.6	5%
Puff Co.	Hungarian	Furniture wholesale	2.2	4%
Wohnung GmbH	Austrian	Furniture retail	2.4	4%

1. Estimate the value of Sweet Home Furniture using multiple-based methods.
2. What other multiples could you apply using the provided data? What further information would you need to enhance the precision of your valuation?
3. What are the pros and cons of using DCF and multiple-based methods in the given case?

An asset-based decision

The WellBuilt Co. is specialised in real estate development and also acts as a main contractor for construction projects. They proved to be successful in a number of cases where they bought the construction sites of liquidated firms to build a resort or an office building. While looking for real estates with good opportunities they came across a former military base with a great view and a lot of old trees, in silent and elegant part of a medium sized city. The owner of the plot is CarryOn Ltd. that offers logistics and transportation services and could purchase the plot at a depressed price some years ago when the current main owner was major of the city. They use the old buildings as storage facilities while the huge open area in the middle of the complex serves as parking area for their trucks. An old office building houses the headquarter of the firm. (It is highly unusual in the industry that companies would own their real estates, rather most of them rent or lease.)

The top management of the WellBuilt Co. decided to make a takeover offer on CarryOn. The current owner is anyway willing to sell as he prepares to restart his political career. They plan to relocate and sell the logistic firm and clear the plot that has currently a book value of 30 million HUF. They expect to pay a yearly fee of 12 million HUF as rental fee for the new headquarter. The fee is predicted to grow by 5 percent. (Rental fees are to be paid in advance in line with the market convention.) WellBuilt expects to sell CarryOn easily at six times of the expected EBITDA (currently 200 million HUF). The logistics firm has an outstanding debt of 120 million HUF.

Based on the plans of WellBuilt Co., 60 percent of plot would be used to set up a resort with apartment buildings and stand-alone homes. They predict to receive 1.25 billion HUF when selling those flats. On the country-road side of the plot, an office building is to be constructed with a useful lifetime of 30 years. (The corporate tax rules allow for a full depreciation over just 20 years.) The net office rental income (before tax and D&A) could reach 60 million HUF yearly (pre-paid). The fee may rise by 5 percent yearly. One fifth of the real estate has to be sold to the local municipality for 10 million HUF (well below the market price) so public roads and street lighting could be established.

1. What is the maximum price to offer for the 100 percent ownership of CarryOn Ltd if the construction works will require two years? Of the 1.1 billion HUF total investment due completely at the end of the works 60 percent would be linked to the apartments. The cost of capital for renting is 10 percent, while that of real estate development amounts to 15 percent, and that of logistics equals 12 percent. The corporate tax rate is 20 percent.
2. What additional problems emerge during this business compared to traditional contraction and development projects?

Together Forever

The Great Tools Ltd. sells and leases out top-quality professional tools in a mid-sized city in Western-Hungary. Their efficiency, profit margins, and cash flows are stable but offer limited upside. However, the owner-manager is not very happy being without no development opportunities. Recently, he had an interesting chat about that at a fair in Budapest with the owner of Wrench Co., a firm in the nearby bigger city.

Slipping their coffee, the two managers decided to merge the two companies and share the gains of a brighter future. They could save on joint marketing, accounting, and procurement mainly done in the capital once every second week. Thus, the joint FCFF may grow by 5 million HUF. On top of that, the growth rate may also increase slightly (by 1 percentage point). This climb could be thanks to offering some jointly purchased high-value special equipment (generators, jackhammers, mini-excavator) for renting that could be stored in one of the two shops and relocated if necessary. Such an offer may boost their competitiveness over smaller players.

The key figures of the two firms before the merger are as follows.

(expected)	FCFF (M HUF)	WACC	g
Great Tools Ltd.	40	10%	3%
Wrench Co.	20	10%	3%

1. Estimate the value of the joint firm.
2. What is the amount of value created by the merger? What part of that would come from the cost-cutting and the faster growth, respectively?
3. Assume that both firms are financed up to 30 percent of their value from a debt costing 10 percent yearly. The corporate tax rate is 20 percent, the industry (or unleveraged) cost of capital is 10.6 percent. If this transaction would be a takeover (not a merger) what would be the maximum price to be paid by the Great Tools for the Wrench? What is the minimum price the owners of the Wrench may accept?
4. Imagine, the Great Tools acquires the Wrench and decides to pass over 40 percent of the synergy generated to the sellers. Half of the price would be paid in cash (this sum is available as a non-operative asset in the buyer firm); the rest will be compensated in ownership rights. What is stake should the seller receive from the new joint company?
5. How and why would the takeover price of Wrench change if Great Tools would expect on a stand-alone FCFF of 60 million HUF?
6. A bank offers a new financing solution for SMEs. Thanks to that the WACC for both firms can be decreased to 9.7 percent. How would that offer affect the value of the firms before and after the transaction? Why would any change happen?

Rainbow Holding

The Rainbow Holding was founded by the equal contribution of five individuals a decade ago. The investment company focuses primarily on private equity opportunities. The firm runs with a minimal amount of operational assets in a rented office with only two of the owners actively taking part in the management. Their portfolio currently includes three companies.

The pure equity-financed Golden Bath Co. manufactures bathroom equipment and prospers pretty well (expected FCFF: 400 million HUF, $g=4\%$). The chief executive officer and the head of the production both own a stake of 47%. The remaining stake is owned by Rainbow as a result of a transaction several years ago where the two managers bought out the former owners and the capital needed for the transaction was borrowed from Rainbow. They purchased a stake and agreed to sell that step-by-step to the two managers based on a predefined plan. The last of the yearly transactions is due in a year. Then Rainbow will leave the firm for a price of 110 million HUF. That step will bring a great relief as the last months brought several heated disputes between the two principal owners regarding the future of the firm, causing serious challenges for the Rainbow management.

Red Pepper Ltd. runs a cosy restaurant in the city centre of a big city. They realised sales of 250 million HUF and paid a dividend of 50 million HUF last year and expect a 3% long term growth. Uniquely for such firms, the company is financed from 500 million HUF debt at a rate of 6%. To upkeep their motivation, the CEO received a stake of 10% while the chef got 5% when the company was purchased 15 years ago. That time the whole kitchen and guest area has been modernised with the aid of Rainbow. Similar restaurants are bought at five times their sales on the market currently. Last month, Rainbow also received an offer at a similar price.

The BlueMotion Co. runs several small cinemas, some of them operating open-air during summertime next to lake Balaton. Currently, the firm that received a considerable amount of capital from Rainbow five years ago saving it from liquidation manages three cinemas. Rainbow owns the firm entirely, and the two managers had a tough time to keep the business running during the last years. They managed to sell off the unneeded central office and paid back the formerly accumulated debt completely. Thus, today, the firm holds only assets (cinema and office equipment, truck) in a value of 30 million HUF. The first long term plan for years finally predicting a stable profit has been finished just some weeks ago. Based on that, perpetuity like FCFE of 15 million HUF will be generated with a yearly growth rate of 5%. Still, it is a crucial issue whether the operating agreements signed with municipalities owning the cinemas can be extended. The current contracts have three years remaining until expiry, and usually, those agreements last for 5 to 10 years.

The unleveraged cost of capital for sanitary equipment manufacturing is 13%, that of hospitality amounts to 11%, while the rate for entertainment and leisure equals to 12%. The risk-free rate is 4%, the corporate tax rate stagnates at 10%.

Recently, one of the managing owners announced that she had to retire due to health issues to spend more time with her grandchildren. She intends to sell her stake to provide some monetary aid to her son, who is currently about building a family house.

1. Estimate the current value of Rainbow.
2. What further information would you need for providing a more precise estimation?
3. What would be your maximum price offered for the 20% stake of Rainbow as an outside investor?
4. How would your results differ if considering the purchase from the point of view the CEO leaving the firm? What about the value for other CEO staying?
5. What do you think the outcome of the transaction would be?

GreatIdea

The venture capital fund RiskIt Co. is about to acquire a stake in GreatIdea Ltd. GreatIdea currently operates without leverage using an equity of 30 M HUF, wholly owned by the founders. Based on the plans of RiskIt, the firm's total would be sold in the best case for 1500 M HUF in 6 years from now.

To convert the existing idea into a product, GoodIdea needs a capital boost of 70 million. The chance of success is 30 percent. In that case, an additional 100 million is required in 2 years to set up mass production. They also expect to receive a 100 million bank loan to cover the purchase of manufacturing equipment. The loan is expected to have a grace period of 5 years. In case of failure, the firm will be liquidated, providing an inflow of cash equal to 90 percent of the initial investment. Inflows would be invested at a yearly rate of 7 percent until the end of the investment horizon. The cost of financing for RiskIt is 20 percent yearly.

- 1) What is the adequate required rate to use for the success case?
- 2) How will the stake of ownership of the investors change because of the equity increases?
- 3) Is it worth investing? What difficulties emerge once we could only sell the firm for 1100 million in 6 years?
- 4) How should the cost of capital, the probability of success, and the return earned in case of failure change so our proposal would still be acceptable from the business point of view?